

Supplement B — Diagnostic Characterisation

category_diagnostic_profiles.csv

34 rows x 10 columns.

Columns: cluster_id, label, domain, size, alternative_model_recovery, resampling_stability, heldout_replication, geometry_separability, heldout_rep_strict, heldout_rep_direction

heldout_replication_summary.csv

34 rows x 11 columns.

Columns: cluster, has_estimate, n_train_enriched, n_test_strict, n_test_lenient, n_test_direction, rep_strict, rep_lenient, rep_dir, axis3_pass, axis3_score

geometry_separability_euclidean.csv

34 rows x 8 columns.

Columns: cluster_id, thematic_label, size, ax4_mean_jaccard, ax4_std_jaccard, ax4_min_jaccard, ax4_max_jaccard, ax4_stable

cluster_id	thematic_label	size	ax4_mean_jaccard	ax4_std_jaccard	ax4_min_jaccard	ax4_max_jaccard	ax4_stable
7	Race start lights	36	0.864485	0.100523	0.666667	1	True
4	Race start launch	42	0.861373	0.129862	0.625	1	True
10	Chicane moves	62	0.82326	0.110916	0.625	1	True
0	DRS battles	205	0.819902	0.0804555	0.6875	0.9375	True
1	Time penalties	118	0.797917	0.122435	0.6	0.933333	True
11	Team radio celebrations	55	0.768748	0.142708	0.533333	0.9375	True
12	Podium finishes	139	0.766865	0.107862	0.590909	0.9375	True
8	Flags signals	118	0.728824	0.151938	0.5	1	True
3	Mirror checks defending	47	0.727672	0.0806435	0.533333	0.8	True
29	Corner specific moves	60	0.716944	0.156349	0.444444	0.933333	True
5	Rain & weather	43	0.716034	0.119302	0.545455	0.933333	True
2	Lights out start	54	0.702917	0.0696232	0.6	0.875	True
6	Grand Prix wins	465	0.686303	0.180638	0.388889	0.933333	True
28	Corner battles early turns	82	0.674589	0.191831	0.25	0.933333	True
9	Formula 1 milestones	134	0.671169	0.146653	0.45	0.933333	True
20	Wing & car damage	61	0.669559	0.12746	0.458333	0.875	True
22	Wheel-to-wheel	71	0.639465	0.163058	0.3125	0.866667	True
16	Safety car deployment	122	0.625658	0.149735	0.375	0.866667	True
19	Race lead holding	115	0.619597	0.249994	0.2	0.933333	True
31	Race leader changes	162	0.605533	0.185789	0.296296	0.866667	True
17	Gap & pace	67	0.601582	0.188062	0.178571	0.875	True
23	Racetrack action	72	0.585244	0.139096	0.384615	0.8	True
15	Tires & compounds	379	0.575147	0.171566	0.352941	0.8125	True
30	Lap timing & fastest lap	136	0.534262	0.202683	0.264706	0.8125	True
21	Braking & lockups	128	0.531874	0.13885	0.333333	0.75	True
13	Championship & title	158	0.491832	0.194068	0.26087	0.9375	False
26	Defending & attacking	110	0.481424	0.188789	0.189189	0.705882	False
25	Indirect driver references	69	0.480582	0.147071	0.243243	0.75	False
24	Car length gaps	60	0.47483	0.180866	0.222222	0.8125	False
14	Pit stops	185	0.446841	0.228526	0.148148	0.882353	False
32	Front position changes	45	0.407461	0.0869679	0.264706	0.588235	False
33	Mid-field positions	122	0.391927	0.115486	0.192308	0.565217	False
18	Team & teammate dynamics	600	0.365429	0.171117	0.166667	0.75	False
27	Overtake battles	762	0.33875	0.0822495	0.166667	0.464286	False

loo_sensitivity.csv

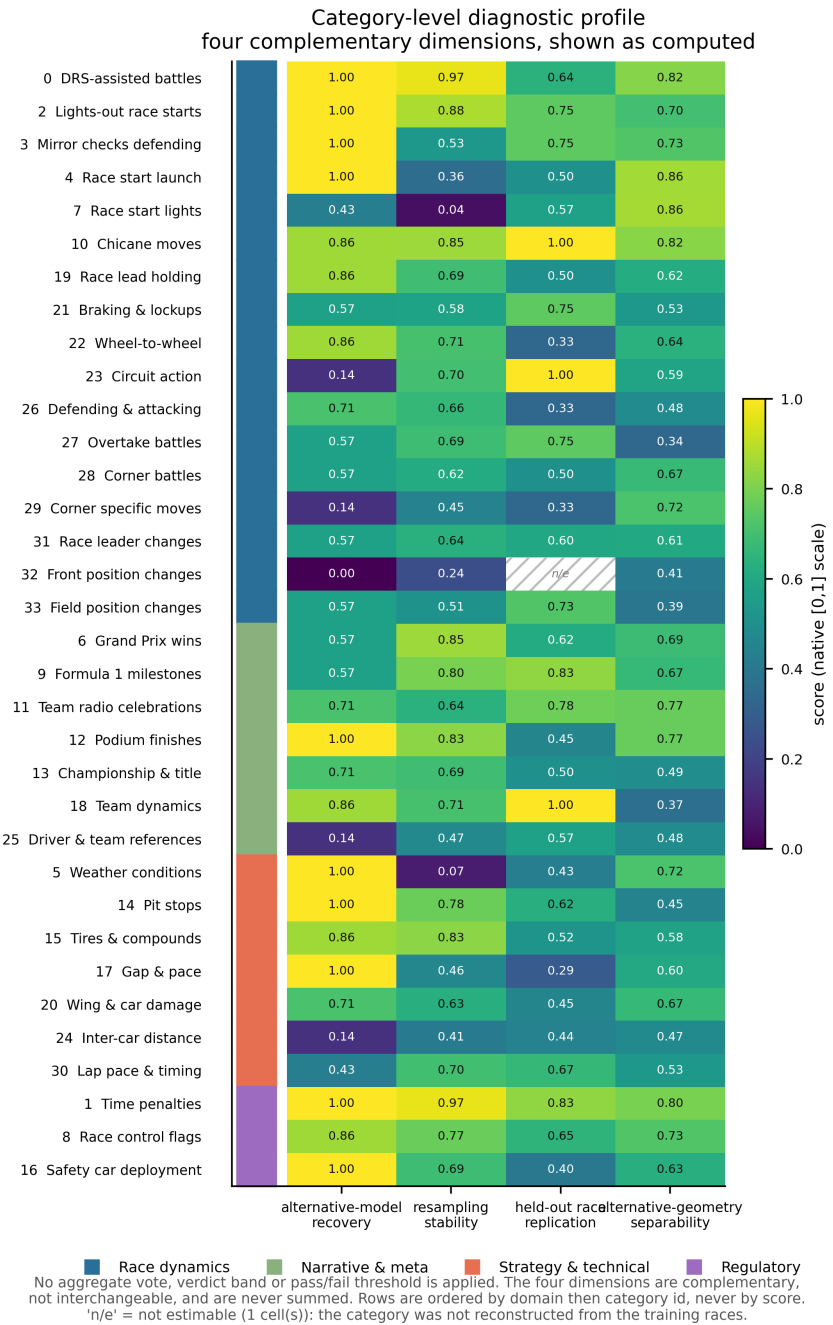
11 rows x 4 columns.

Columns: removed, majority_count, delta, abs_delta

removed	majority_count	delta	abs_delta
(none - all judges)	28	0	0
enc_all	28	0	0
enc_bge	27	-1	1
enc_gte	28	0	0
kmeans	28	0	0
ward	28	0	0
optics	28	0	0
leiden	27	-1	1

removed	majority_count	delta	abs_delta
res_mcs17	27	-1	1
res_mcs52	27	-1	1
seed_umap2	27	-1	1

Figures



Category-level diagnostic profile across four complementary dimensions, shown as computed. No aggregate vote or verdict band.